
Macro and Option Information in Systematic FX Carry

Predicting the distribution of carry returns, and turning that prediction into exposure

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Week 6 Research Update

THE QUESTION

Carry harvests a premium and pays for it in risk-off episodes. Can **macro state variables** and **option-implied information** say in advance when to be smaller, and what to hold?

THE DESIGN

20-currency G10 and EM panel, 234 months, 2007–2026. Chronological discovery, validation and **frozen evaluation** samples; HAC inference; multiple-testing control; point-in-time audits.

Source of record: 08_comprehensive_carry_strategy_optimization(6).ipynb; Notebook 07 supplies the inherited option architectures. Every number is machine-checked against the notebook's saved outputs — see the fact-check register in Appendix A9.

The problem, and the three questions that follow

Carry is payment for bearing a specific risk, so the interesting question is when to bear less of it

THE ECONOMIC PROBLEM

A pooled G10 and EM carry book is long high-yield currencies against low-yield funders. It earns steadily, then returns several years of gains in a few months: gross Sharpe **0.58** with a **-19.0%** maximum drawdown over 2007–2026.

Losses cluster in identifiable states. That is what makes conditioning information potentially valuable.

DISCIPLINE

Every answer is reported with its sample, its inference method, and what survives multiple-testing correction. Where nothing survives, the slides say so.

RQ1 — PREDICTABILITY IN WHICH MOMENT?

Do formation-time macro conditions predict the *mean*, the *magnitude*, or the *tail* of next-month carry returns? Three questions, three different answers.

RQ2 — DOES IT CONVERT INTO A BETTER PORTFOLIO?

Can a macro risk signal become a defensive gross-exposure rule that improves risk-adjusted performance, rather than a directional timing model?

RQ3 — IS MACRO INFORMATION INCREMENTAL?

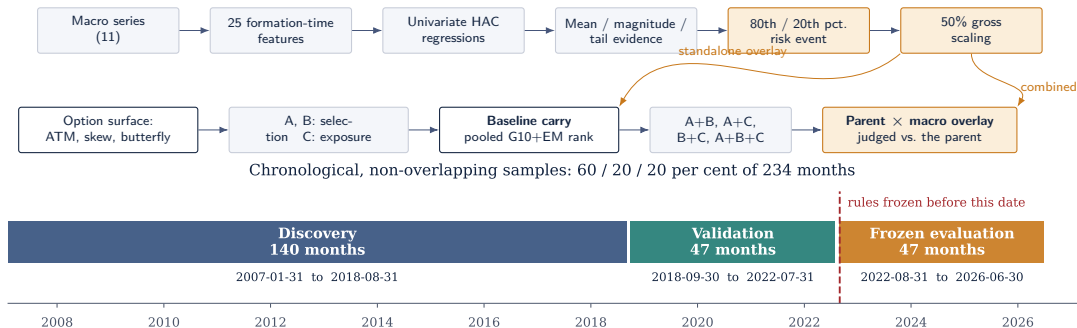
Option signals already improve carry selection and sizing. Does macro conditioning add anything *after* that, judged against the option-conditioned parent rather than against baseline?

WHY THE FRAMING MATTERS

A signal that fails to forecast the sign of returns can still forecast their size. The first justifies changing the ranking; the second justifies changing the scale. Most of this study turns on keeping those two claims apart.

Research architecture

Two information sources, two distinct portfolio actions, one incremental-value test



WHY CHRONOLOGICAL, NOT RANDOM

Macro states, volatility and carry returns are persistent. A random split leaks neighbouring months across the boundary and lets a rule learn from its own future.

HOW EACH SAMPLE IS USED

Evidence labels come from **discovery and validation only**. The 47-month evaluation window is robustness evidence, never the headline — far too short to resolve a Sharpe difference.

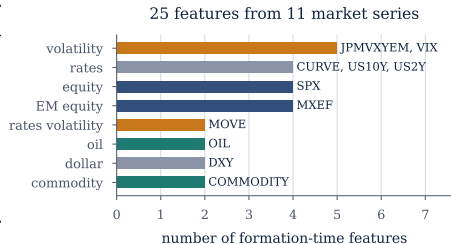
The macro universe: eleven series, organised by economic role

Volatility, dollar and rates, equity, commodities — 25 formation-time features

Category	Series	#	Transforms	Risky	Economic reading
EM equity	MXEF	4	$r1m$, $mom3m$, $dd6m$, $r1m(tail)$	low	EM equity
commodity	COMMODITY	2	$r1m$, $mom3m$	low	broad commodity index
dollar	DXY	2	$r1m$, $mom3m$	high	broad dollar
equity	SPX	4	$r1m$, $mom3m$, $dd6m$, $r1m(tail)$	low	US equity
oil	OIL	2	$r1m$, $mom3m$	low	front WTI
rates	CURVE	2	level, $\Delta 1m$	high	US 2s10s slope
rates	US10Y	1	$\Delta 1m$	high	long US yield
rates	US2Y	1	$\Delta 1m$	high	front-end US yield
rates vol	MOVE	2	level, $\Delta 1m$	high	US rates volatility
volatility	JPMVXYEM	2	level, $\Delta 1m$	high	EM FX implied volatility
volatility	VIX	3	level, $\Delta 1m$, $\Delta 3m$	high	US equity-implied volatility

Full 25-feature table with risk signs and evidence flags is in Appendix A1. Source:

08v6_macro_feature_audit.parquet.



THE ORGANISING IDEA

Not eleven independent bets but four readings of one state: **how much risk is priced, and how much has been realised.**

WHY EM EQUITY AND OIL

The long leg is predominantly EM, so MXEF and oil sit closest to the exposures the book actually carries — a prior worth holding before any p -value.

Feature engineering and point-in-time construction

A transformation, a historical z-score, and a risk direction — all evaluated at the formation month end

1. TRANSFORMATION

Levels; one- and three-month differences for yields and volatility indices; one-month log returns and three-month log momentum for prices; six-month drawdown $P_t / \max_{s \in [t-5, t]} P_s - 1$ for equity.

2. HISTORICAL NORMALISATION

$$z_{j,t} = \frac{X_{j,t} - \bar{X}_{j,[t-60, t-1]}}{S_{j,[t-60, t-1]}}$$

Trailing 60 months, minimum 24 observations, both moments **lagged one month**: the statistic at t uses only data observable strictly before t .

3. RISK DIRECTION

$$\tilde{z}_{j,t} = \rho_j z_{j,t}, \quad \rho_j = \begin{cases} +1 & \text{higher is riskier} \\ -1 & \text{lower is riskier} \end{cases}$$

13 features are higher-is-riskier (volatility, dollar, yields, curve), 12 lower-is-riskier (equity, commodity, oil).

WHY THE SIGN CONVENTION EARNS ITS PLACE

Without ρ_j , “high VIX is risky” and “low oil is risky” need opposite thresholds and opposite expected signs everywhere. With it, one rule and one expected sign cover all 25 features.

WHAT POINT-IN-TIME MEANS HERE

Three quantities are lagged together: the normalising mean, its standard deviation, and the percentile threshold that later defines a risk event. The audits report **zero future-horizon violations**.

ONE CAVEAT, STATED UP FRONT

SPX_tail_return_1m and MXEF_tail_return_1m are *numerically identical* to the matching _return_1m features: the label describes the intended reading, not a different transformation. The notebook flags both as duplicate-equivalent and **excludes them from the paired comparison**. Treating them as independent findings would overstate the breadth of the result.

Regression framework

Univariate, one macro predictor at a time, Newey–West inference, explicit multiple-testing control

TWO TIMINGS, TWO MEANINGS

$$\text{Predictive: } Y_{t+1} = \alpha + \beta X_t + \varepsilon_{t+1}$$

$$\text{Contemporaneous: } Y_{t,t+1} = \alpha + \beta X_{t,t+1} + \varepsilon_{t,t+1}$$

The first uses only what is known at formation. The second relates the return to the macro move during the holding period: a mechanism, never a tradable signal.

ESTIMATION

OLS with intercept, one predictor at a time — there is no 25-variable regression. HAC/Newey–West with **3 lags**, minimum 18 observations; panel variants cluster by month.

MULTIPLE TESTING

Benjamini–Hochberg and Holm within each (timing, sample, dependent variable) family; nominal p -values always appear beside their q .

A GRID, NOT A SINGLE TEST

Dimension	Levels
Macro feature	25
Predictor form	continuous z ; binary risk-event indicator
Timing	predictive; contemporaneous
Component	full book; long leg; short leg; currency panel
Outcome	mean; magnitude; tail
Sample	full; discovery; validation; evaluation

THREE OUTCOME FAMILIES

$$\underbrace{r_{t+1}}_{\text{mean}} \quad \underbrace{|r_{t+1}|, r_{t+1}^2}_{\text{magnitude}} \quad \underbrace{\mathbf{1}\{r_{t+1} < 0\}, \mathbf{1}\{r_{t+1} \leq q_t^{5\%}\}}_{\text{tail}}$$

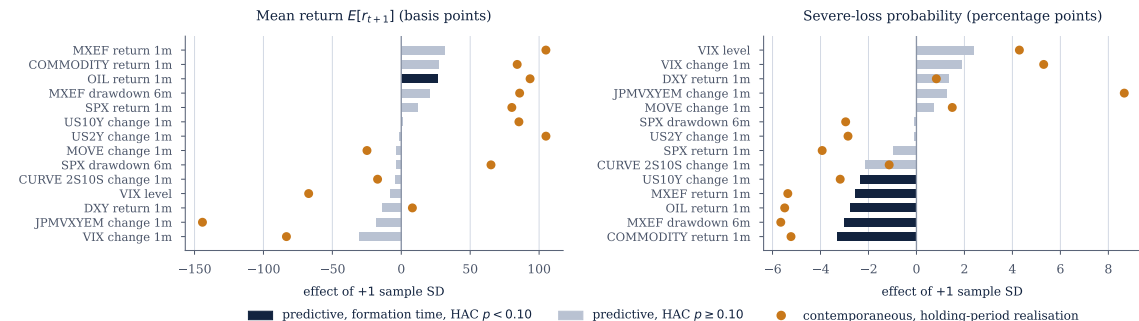
A severe loss is a month in the lower 5% of the book's own trailing 60-month distribution, threshold lagged like everything else.

COMPARABILITY RULE

Coefficients are reported as **standardised economic effects**: the change in the outcome per +1 regression-sample standard deviation of the predictor, in basis points for returns and percentage points for probabilities. Comparable *within* an outcome, never across outcomes.

Predictive versus contemporaneous effects

Fourteen matched pairs: what a macro variable says at formation, against what its holding-period move explains after the fact



HOW TO READ IT

Bars are formation-time predictive effects, dots the matched holding-period realisation, both per +1 sample standard deviation.

WHAT IT SHOWS

Dots are almost always **far larger** than bars: macro moves explain much of carry returns *ex post* and little *ex ante*.

WHY IT MATTERS

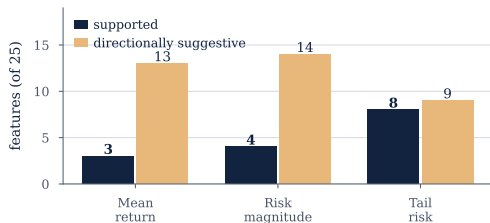
This is why the strategy work targets **exposure** rather than direction: we can identify the states that hurt long before we can forecast the return.

Only the 14 semantically matched pairs are plotted; 9 predictive features have no holding-period counterpart and 2 are duplicate-equivalent. 08v6_lead_concurrent_plot_pair_audit.parquet.

Where the predictive evidence actually is

Broad and consistent for tail risk, present for magnitude, real but narrow for mean returns

Evidence is broadest for tail risk



WHAT "SUPPORTED" REQUIRES

The economically expected sign in **both** discovery and validation, valid inference in both, and clear discovery significance ($p < 0.10$ or BH $q < 0.20$). The frozen evaluation window plays no part in the label. The full 25-feature grid is in Appendix A1b.

Full-sample standardised coefficients from 08v6_predictive_portfolio_regressions.parquet; support labels from 08v6_predictive_evidence_map.parquet. Recall that MXEF_tail_return_1m duplicates MXEF_return_1m.

Outcome	Feature	Std. coef.	HAC t	HAC p	BH q
Mean return	MXEF_return_1m	+31.44	+1.51	0.130	0.835
	MXEF_tail_return_1m	+31.44	+1.51	0.130	0.835
	COMMODITY_momentum_3m	+24.76	+1.67	0.094	0.835
Risk magnitude	VIX_level	+39.66	+2.95	0.003	0.027
	JPMVXYEM_change_1m	+24.52	+2.61	0.009	0.065
	MXEF_momentum_3m	-41.77	-3.39	0.001	0.018
	MXEF_drawdown_6m	-41.73	-3.61	0.000	0.015
Tail risk	JPMVXYEM_change_1m	+1.27	+1.07	0.285	0.619
	MXEF_return_1m	-2.55	-2.37	0.018	0.147
	MXEF_momentum_3m	-2.79	-2.49	0.013	0.147
	MXEF_drawdown_6m	-3.02	-2.42	0.015	0.147
	MXEF_tail_return_1m	-2.55	-2.37	0.018	0.147
	COMMODITY_return_1m	-3.33	-2.10	0.036	0.224
	COMMODITY_momentum_3m	-2.52	-2.12	0.034	0.224
	OIL_return_1m	-2.75	-1.89	0.058	0.266

THE CONCLUSION, STATED PRECISELY

Macro information is **not** silent on mean returns — three features clear the bar. But breadth and cross-sample consistency sit with **adverse tail outcomes**, and the EM-equity, commodity and oil complex supplies most of it. That tells us which moment to trade, not whether to trade.

From a continuous signal to a portfolio rule

A risk event is a tail of the signal's own history; the response is a halving of exposure, not a change of view

1. DEFINE THE EVENT

$$E_{j,t} = \begin{cases} 1\{z_{j,t} \geq Q_{j,[t-60, t-1]}^{80}\}, & \rho_j = +1, \\ 1\{z_{j,t} \leq Q_{j,[t-60, t-1]}^{20}\}, & \rho_j = -1. \end{cases}$$

The percentile is rolling, historical and lagged, exactly like the z-score. A feature is risky relative to its own recent history, not to a level chosen with hindsight.

2. SCALE THE PARENT

$$w_{i,t}^{\text{macro}} = m_t w_{i,t}^{\text{parent}}, \quad m_t = \begin{cases} 0.50 & E_t = 1, \\ 1.00 & E_t = 0. \end{cases}$$

One scalar multiplies every weight, so composition is untouched and only size changes; undefined events default to $m_t = 1$. With $\text{Gross}_t = \sum_i |w_{i,t}|$ and a baseline fixed at 2.00, an active month runs at 1.00.

WHY THIS IS THE RIGHT SHAPE OF RESPONSE

The evidence says macro predicts *when losses cluster*, not *which way the market goes*. A rule that changed the ranking would be trading the mean; a rule that changes the scale trades the variance and the tail. Only the second follows from Section 1.

DESIGN CHOICES, AND THEIR COST

The 80th/20th percentile makes events roughly a fifth of months (46 of 234 for oil): frequent enough to matter, rare enough to leave the premium intact. The 50% cut is deliberately blunt.

Zero scaling is carried through as a sensitivity and usually looks slightly better on Sharpe. It is not the primary rule: taking the book to zero on one indicator is a stronger claim than the evidence supports.

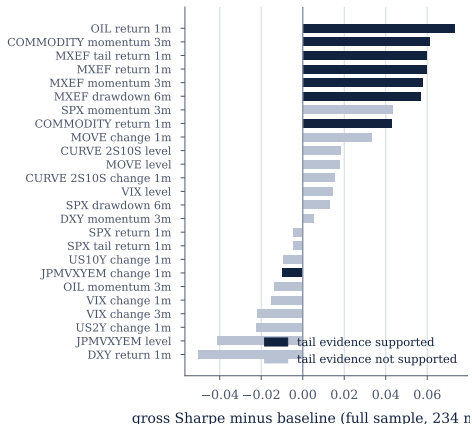
WHAT THE RULE CANNOT DO

It cannot help in months it does not fire, it cannot tell a bad month from a bad month *for this book*, and it pays for every false alarm. The next two slides quantify all three.

Every macro feature as a standalone overlay on baseline carry

25 rules, one construction, no parameter search

Every feature as a 50% de-risking overlay on base



50% overlay on	Act.	Ret.	Vol.	SR	ΔSR	ES	Gr.
Baseline carry	—	+5.24%	9.07%	+0.58	—	-6.47%	2.00
OIL_return_1m	20%	+5.23%	8.03%	+0.65	+0.073	-5.30%	1.80
COMMODITY_momentum_3m	22%	+5.19%	8.12%	+0.64	+0.061	-5.69%	1.78
MXEF_tail_return_1m	17%	+5.29%	8.30%	+0.64	+0.060	-5.72%	1.83
MXEF_return_1m	17%	+5.29%	8.30%	+0.64	+0.060	-5.72%	1.83
MXEF_momentum_3m	20%	+5.11%	8.03%	+0.64	+0.058	-5.59%	1.80
MXEF_drawdown_6m	23%	+4.97%	7.83%	+0.63	+0.057	-5.24%	1.77
SPX_momentum_3m	20%	+5.07%	8.15%	+0.62	+0.043	-5.74%	1.80
COMMODITY_return_1m	18%	+5.21%	8.38%	+0.62	+0.043	-5.88%	1.82
MOVE_change_1m	18%	+5.02%	8.22%	+0.61	+0.033	-5.62%	1.82
CURVE_2S10S_level	13%	+5.10%	8.56%	+0.60	+0.018	-5.79%	1.87

READING THE LEADERBOARD

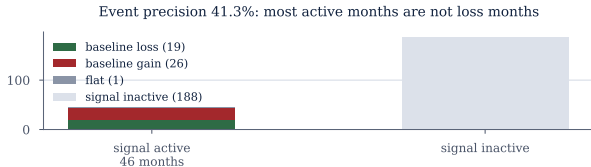
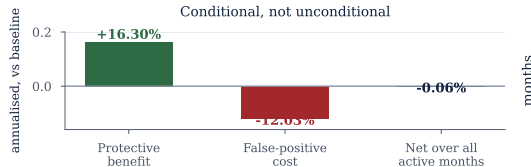
Same rule, same book, only the trigger changes. Dark bars mark features whose *tail* evidence was labelled supported before any portfolio was built, so the ordering is not purely a backtest ranking.

THE PATTERN, AND THE CAUTION

Improvements arrive with **lower average gross** (1.5–1.8 against 2.00) and *unchanged* maximum drawdown: the de-risking trims ordinary variance rather than catching the worst episode. And these are 25 rules on the same 234 months — a distribution of outcomes under one design, not 25 independent discoveries.

Case study: the oil overlay is protection, not alpha

The most instructive result in the study, and the one most easily misread



Statistic	Value	Reading
Full-sample gross Sharpe	+0.65	baseline 0.58
Frozen-evaluation gross Sharpe	+1.59	baseline over the same 47 months 1.78
Average gross exposure	1.80	baseline 2.00
Expected shortfall (5%)	-5.30%	improvement of +1.17 points
Signal-active months	46 of 234	19.7% activation
Event precision	41.3%	19 loss months of 46 active
Conditional protective benefit	+16.30%	active <i>and</i> baseline-loss months
Conditional false-positive cost	-12.03%	active <i>and</i> baseline-gain months
Net over all active months	-0.06%	the two conditional effects cancel

WHAT THE 16.3% IS, AND IS NOT

It annualises a mean monthly relative return of 1.36% computed **only over months when the signal was active and baseline then lost money**: a conditional attribution statistic, **not** a 16.3% return improvement and **not** alpha. In active months where baseline *gained*, being half-sized cost 12.03%.

NETTING THE TWO

At 41.3% precision the two cancel to **-0.06%** across all active months. Sharpe still rises from 0.58 to 0.65 — by removing variance, not adding return: a legitimate thing for a macro signal to be, just not a timing model.

Composite macro signals: does breadth beat the best single feature?

Five pre-defined composites, and a direct answer

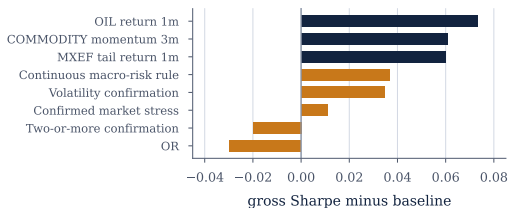
Composite	Activation rule	Active	Sharpe	Δ SR	Gross
OR	any one of the 25 feature events is active	72%	+0.55	-0.030	1.28
Two-or-more confirmation	at least two of the 25 feature events are active	59%	+0.56	-0.020	1.41
Volatility confirmation	at least one volatility-family event (VIX, JPMVXYEM, MOVE)	46%	+0.61	+0.035	1.54
Confirmed market stress	at least one event among volatility, equity, dollar and rates families	65%	+0.59	+0.011	1.35
Continuous macro-risk rule	$\text{MacroRisk}_t \geq$ its own historical 80th percentile	19%	+0.61	+0.037	1.81
Baseline carry	—	0%	+0.58	—	2.00

HOW MACRORISK IS BUILT

$$\text{MacroRisk}_t = \frac{1}{25} \sum_{j=1}^{25} \rho_j z_{j,t},$$

the average direction-normalised z-score, thresholded at its own rolling 80th percentile. It is the only composite that uses signal *strength* rather than a count of binary events.

Single features versus composites



THE ANSWER IS NO

The best composite adds **+0.037** Sharpe against baseline; the best single features add **+0.06 to +0.07**.

WHY BREADTH DISAPPOINTS HERE

This is **feature-level** breadth, not asset-class breadth: four of 25 features are MXEF transformations and five are volatility indices, so an “any-of-25” rule fires repeatedly on the same state.

The inherited option-conditioned architectures

Notebook 07 in one slide: two ways to use the option surface, and they are economically different

Label	Notebook 07 strategy	Economic role	Gross exposure
A	G10-aware ATM conditioning ($\lambda_{G10}=0.5$, $\lambda_{EM}=0$)	selection on ATM volatility	fixed, ≈ 2.0
B	Butterfly-conditioned carry ($\lambda_B = -0.5$)	selection on smile curvature	fixed, ≈ 2.0
C	Joint option-risk scaling, variable-gross ($\gamma=0.5$)	exposure control, same basket	variable, ≤ 2.0

SELECTION: CHANGE WHAT YOU HOLD

The score is carry times a bounded option factor, so the sign of carry survives and only the ordering moves:

$$\text{Score}_{i,t} = C_{i,t} \text{ clip}\left(1 + \lambda z_{i,t}^X, 0.25, 2.00\right).$$

A sets $\lambda = 0.5$ on ATM volatility for G10 and 0 for EM, asymmetric because the Notebook 06 carry $\times$ ATM interaction was a G10 result.

B sets $\lambda_B = -0.5$ on the butterfly, so high smile curvature compresses carry. Gross stays at 2.00.

EXPOSURE CONTROL: CHANGE HOW MUCH YOU HOLD

C keeps the baseline basket and shrinks positions that look expensive to hold:

$$m_{i,t} = \text{clip}\left(\frac{1}{1+\gamma \sum_X \frac{1}{\max(z_{i,t}^X, 0)}}, 0.25, 1.00\right),$$

$\gamma = 0.5$ over ATM volatility, skew and butterfly. Under **variable gross** the legs are rescaled to $G_t = \min(L_t, S_t)$, keeping the book net-neutral as gross falls below 2.

WHY THE DISTINCTION DRIVES EVERYTHING AFTER THIS

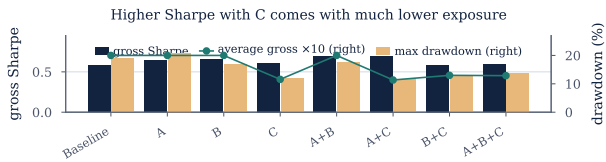
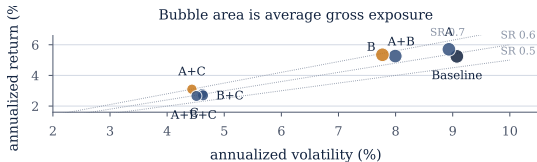
Selection and exposure can post the same Sharpe for completely different reasons. Any comparison that ignores average gross credits risk reduction as if it were skill.

Option z-scores are monthly within-asset-class standardisations from Notebook 06; Notebook 08 reproduces the Notebook 07 weights, returns, turnover and costs to within 10^{-12} before using them.

Core option architectures: what improves, and why

Eight books, same 234 months — ranked, but read alongside average gross

Arch.	Mechanism	Full sample, 234 months								Eval.
		Ret.	Vol.	Sharpe	Δ SR	Max DD	ES	Gross	Turn.	Sharpe
Baseline	baseline carry	+5.24%	9.07%	+0.58	+0.000	-19.00%	-6.47%	2.00	0.18	+1.78
A	selection improvement	+5.70%	8.93%	+0.64	+0.060	-20.81%	-6.14%	2.00	0.23	+1.96
B	selection improvement	+5.26%	7.99%	+0.66	+0.080	-16.89%	-5.14%	2.00	0.29	+1.87
C	variable-gross risk control	+2.81%	4.61%	+0.61	+0.030	-11.80%	-3.08%	1.16	0.18	+1.64
A+B	selection improvement	+5.35%	7.77%	+0.69	+0.110	-17.51%	-4.96%	2.00	0.33	+1.69
A+C	selection + variable-gross risk control	+3.10%	4.43%	+0.70	+0.121	-11.57%	-2.92%	1.14	0.21	+1.94
B+C	selection + variable-gross risk control	+2.70%	4.63%	+0.58	+0.005	-12.65%	-3.06%	1.30	0.25	+1.72
A+B+C	selection + variable-gross risk control	+2.66%	4.51%	+0.59	+0.011	-13.86%	-2.96%	1.29	0.27	+1.49



TWO ROUTES TO ONE SHARPE

A+B reaches 0.69 at gross 2.00 on 5.35% against baseline's 5.24%: *selection*. **A+C** reaches 0.70 at gross **1.14** on 3.10% — from the denominator.

READING A+C HONESTLY

Halving exposure leaves Sharpe unchanged by construction; A+C beats that, so the scaling carries information — but levering back to gross 2.00 would widen the drawdown.

THE DECOMPOSITION

Everything containing **C** sits in the low-volatility, low-gross cluster. **Selection moves the numerator, exposure control the denominator.**

Combining macro and option information

The construction, and the comparison that makes it a real test

CONSTRUCTION

The macro overlay multiplies the parent's own weights:

$$w_{i,t}^{\text{comb}} = m_t w_{i,t}^{\text{parent}}, \quad m_t \in \{0.5, 1\}.$$

For a fixed-gross parent such as A+B this takes gross from 2.00 to 1.00. For a **variable-gross** parent such as A+C it halves whatever the parent was already running.

THE COMPARISON THAT MATTERS

A combined strategy must be judged against **its own parent**. Against baseline almost every combination looks like an improvement — but so does the parent alone, and the macro layer would be taking credit for the option layer's work.

METRICS

Δ Sharpe, Δ return, Δ max drawdown, Δ evaluation Sharpe, and the change in average gross — the last always reported, so exposure reduction is never mistaken for selection skill.

Parent	Overlay	Gross Sharpe			vs. parent		Avg. gross
		Par.	Comb.	Δ	Ret.	MDD	
Baseline	OIL	+0.58	+0.65	+0.073	-0.01%	+0.00%	2.00 → 1.80
A	MXEF	+0.64	+0.71	+0.071	+0.12%	+0.00%	2.00 → 1.83
A	OIL	+0.64	+0.70	+0.060	-0.10%	+0.00%	2.00 → 1.80
B	OIL	+0.66	+0.72	+0.060	-0.10%	+0.00%	2.00 → 1.80
A+B	OIL	+0.69	+0.74	+0.048	-0.16%	+0.00%	2.00 → 1.80
A+C	OIL	+0.70	+0.73	+0.028	-0.17%	+0.00%	1.14 → 1.02
A+C	MXEF	+0.70	+0.74	+0.038	-0.07%	+0.00%	1.14 → 1.03
A+B+C	OIL	+0.59	+0.58	-0.008	-0.23%	+0.33%	1.29 → 1.16

WHAT THE ROWS SAY

Adding the oil overlay to **baseline** carry is worth +0.073 Sharpe; adding it to **A+C** is worth +0.028 and cuts average gross from 1.14 to 1.02. The better a parent already manages its exposure, the less a second rule has left to do.

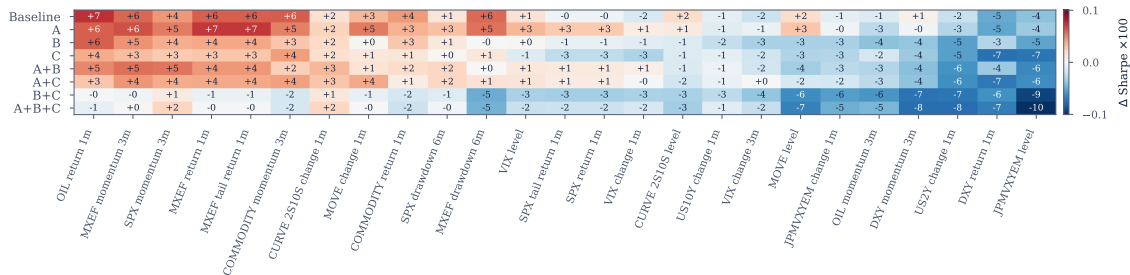
DIMINISHING RETURNS

Both layers detect the same stressed state through different instruments. Where they overlap, the second mostly adds turnover.

Incremental value across the full grid

Eight option parents × 25 macro features, each cell measured against its own parent

Incremental value of a 50% macro overlay, measured against each option parent (not against baseline)



HOW TO READ IT

Red improves the parent, blue worsens it; values are Sharpe changes × 100. Columns are sorted by average effect, so the left edge is the EM-equity, commodity and oil complex.

TWO CLEAR PATTERNS

Columns dominate **rows**: whether a feature helps is mostly a property of the feature, and it lines up with the tail evidence. Rows fade downwards — the bottom parents already contain C.

THE HEADLINE

Of 200 pairs, **89 improve** their parent and **111 worsen** it. Macro on top of options is **not** systematic; it is selective, and concentrated where the evidence pointed beforehand.

08v6_macro_extension_incremental_results.parquet. This is a pre-defined cross-product, not a search: every parent is crossed with every feature and all 200 outcomes are reported.

Risk periods and crises

Two definitions of “a bad month”, kept separate on purpose

Cumulative return in five pre-defined crisis windows (%)

	GFC 2008-09	Taper 2013	EM/China 2015-16	Covid 2020	USD/Rates 2022
Baseline	-9.2	-4.8	-4.6	-7.2	+12.7
A	-10.3	-4.8	-3.7	-6.8	+12.3
B	-9.4	-5.9	-1.8	-6.3	+13.8
A+B	-11.4	-5.9	-0.5	-5.4	+14.1
A+C	-6.8	-3.8	-1.2	-3.1	+8.1
OIL 50% overlay	-9.2	-5.3	-1.5	-1.4	+10.0

Cell values are total returns over each fixed window; the underlying table, with average gross exposure per strategy, is in Appendix A7c.

FIXED CRISES VERSUS RULE-DEFINED PERIODS

Fixed crisis windows are five calendar episodes chosen with hindsight: they test whether a strategy survived history. **Rule-defined active months** are the months a signal fired: they test whether the rule does what it claims. Never combined.

WHAT THE GRID SHOWS

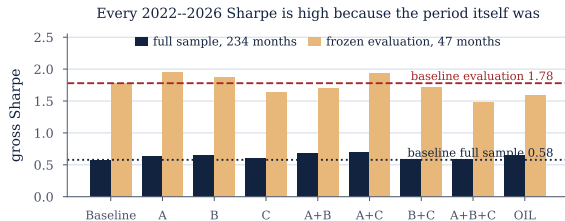
Selection architectures do *not* reliably help in a crisis: A+B lost more than baseline in both the GFC and the taper tantrum. The exposure-based routes do. **A+C** roughly halves the GFC and Covid losses; the **oil overlay** cuts Covid from -7.2% to -1.4% but leaves the GFC untouched — it did not fire in 2008.

THE GENERAL LESSON

Protection is episode-specific. A signal built on one transmission channel protects against shocks that travel through that channel and is blind to the rest. That is an argument for describing what a rule does, not for calling it a hedge.

Robustness, and what the statistics will not support

The four screens that decide how much of this is believable

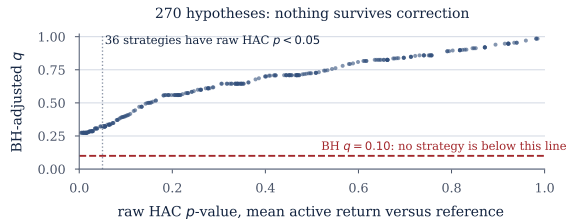


1. FROZEN EVALUATION

Every 2022–2026 Sharpe looks spectacular because the **baseline itself** posts 1.78 over those 47 months. Against that bar A+C is +0.16, A+B −0.09, the oil overlay −0.19.

3. POINT-IN-TIME

Zero future-horizon violations across the macro, option and strategy lineages. But verification is uneven — 50 strategies have source information too thin to check beyond lineage — so calling all 271 independently verified would be false. Appendix A7b.



2. MULTIPLE TESTING

36 of 270 strategies have a raw HAC $p < 0.05$ on their mean active return. After Benjamini–Hochberg, **none** has $q < 0.10$ — and that is why Tier 1 is empty.

4. COSTS, AND WHAT THE COUNTS MEAN

Selection books turn over more (0.33 for A+B against 0.18), so their advantage erodes fastest; 1–50 bp sensitivity in Appendix A6. The universe is 271 canonical strategies after de-duplicating 323 aliases — **not** 271 independent hypotheses.

Conclusions

Stated at the strength the evidence supports

1. MACRO PREDICTABILITY IS REAL, AND IT IS ABOUT RISK

Across 25 point-in-time features, **8** carry supported evidence on severe-loss probability, **4** on return magnitude and **3** on mean returns. Macro is not silent on the mean — but breadth, cross-sample consistency and economic coherence all sit with the **tail**.

2. THAT MAKES MACRO A SIZING TOOL, NOT A TIMING TOOL

The oil overlay is the clean case: a large conditional benefit in signal-active loss months (**+16.3%** annualised conditional), most of it paid back in false positives (**−12.03%**), netting **−0.06%** across active months at 41.3% precision. Full-sample Sharpe still improves from 0.58 to 0.65 — through variance reduction, not added return.

3. OPTION INFORMATION WORKS THROUGH TWO CHANNELS

A+B improves Sharpe to 0.69 at unchanged gross 2.00: selection. **A+C** reaches 0.70 at gross 1.14 with a **−11.6%** drawdown: exposure. Both are useful; conflating them credits de-risking as skill.

4. MACRO ADDS SELECTIVELY AFTER OPTIONS, NOT SYSTEMATICALLY

Of 200 parent-feature combinations, **89 improve** the parent and **111 do not**, and the winners concentrate in the same EM-equity, commodity and oil features the regressions had already flagged. Parents that already scale their own exposure gain least.

THE GOVERNING CAVEAT

No strategy survives multiple-testing correction and the Tier 1 shortlist is empty by construction. The defensible claim is about *mechanism* — which moment macro predicts and which portfolio action that justifies — not about a validated tradable edge.

Appendix

Full feature table, regression detail, leaderboards, cost sensitivity, taxonomy and the fact-check register

A1 — The 25 macro features

Support flags: M mean, G magnitude, T tail; a dot means not supported

Series	Category	Feature id	Transformation	Risk sign	Support
VIX	volatility	VIX_level	level at formation month end	+1	.G.
VIX	volatility	VIX_change_1m	1-month difference	+1	...
VIX	volatility	VIX_change_3m	3-month difference	+1	...
JPMVXYEM	volatility	JPMVXYEM_level	level at formation month end	+1	...
JPMVXYEM	volatility	JPMVXYEM_change_1m	1-month difference	+1	.GT
MOVE	rates volatility	MOVE_level	level at formation month end	+1	...
MOVE	rates volatility	MOVE_change_1m	1-month difference	+1	...
DXY	dollar	DXY_return_1m	1-month log return	+1	...
DXY	dollar	DXY_momentum_3m	3-month log return	+1	...
SPX	equity	SPX_return_1m	1-month log return	-1	...
SPX	equity	SPX_momentum_3m	3-month log return	-1	...
SPX	equity	SPX_drawdown_6m	price / 6-month rolling max - 1	-1	...
SPX	equity	SPX_tail_return_1m	1-month log return (lower-tail label)	-1	...
MXEF	EM equity	MXEF_return_1m	1-month log return	-1	M.T
MXEF	EM equity	MXEF_momentum_3m	3-month log return	-1	.GT
MXEF	EM equity	MXEF_drawdown_6m	price / 6-month rolling max - 1	-1	.GT
MXEF	EM equity	MXEF_tail_return_1m	1-month log return (lower-tail label)	-1	M.T
US2Y	rates	US2Y_change_1m	1-month difference	+1	...
US10Y	rates	US10Y_change_1m	1-month difference	+1	...
CURVE	rates	CURVE_2S10S_level	level at formation month end	+1	...
CURVE	rates	CURVE_2S10S_change_1m	1-month difference	+1	...
COMMODITY	commodity	COMMODITY_return_1m	1-month log return	-1	..T
COMMODITY	commodity	COMMODITY_momentum_3m	3-month log return	-1	M.T
OIL	oil	OIL_return_1m	1-month log return	-1	..T
OIL	oil	OIL_momentum_3m	3-month log return	-1	...

Risk sign +1 means higher is riskier. Transformations are applied to month-end observations and converted to a trailing 60-month, one-month-lagged z-score. 08v6_macro_feature_audit.parquet

joined to 08v6_predictive_evidence_map.parquet.

A1b — The full evidence grid

All 25 features against the three outcome families

Predictive evidence by outcome
S = supported, s = directionally suggestive

	Mean $E[r_{t+1}]$	Magnitude $E[r_{t+1}]$	Tail $P(\text{severe loss})$
COMMODITY momentum 3m	S	s	S
JPMVXYEM change 1m	s	S	S
MXEF drawdown 6m	s	S	S
MXEF momentum 3m		S	S
MXEF return 1m	S	s	S
MXEF tail return 1m	S	s	S
COMMODITY return 1m	s	s	S
OIL return 1m	s	s	S
VIX level		S	s
CURVE 2S10S change 1m	s		
CURVE 2S10S level			
DXY momentum 3m	s	s	s
DXY return 1m	s		s
JPMVXYEM level		s	
MOVE change 1m	s	s	
MOVE level		s	s
OIL momentum 3m	s	s	s
SPX drawdown 6m		s	
SPX momentum 3m		s	s
SPX return 1m	s		s
SPX tail return 1m	s		s
US10Y change 1m			
US2Y change 1m			
VIX change 1m	s	s	s
VIX change 3m	s	s	

HOW TO READ IT

One row per feature, one column per outcome family. S marks supported evidence — expected sign in discovery and validation with valid inference and clear discovery significance. s marks directionally suggestive: the right sign without the significance condition. Blank means neither.

THE SHAPE OF THE TABLE

The tail column is densest, and the rows that fill are the EM-equity, commodity and oil transformations. Volatility features concentrate in magnitude; dollar, rates and US-equity features are largely suggestive-only.

A CAUTION ABOUT COUNTING

Rows are not independent: four are MXEF transformations of the same index and two of the 25 are numerically duplicated tail-return labels. The count of supported cells overstates the number of distinct findings.

A2 — Regression specification detail

ESTIMATOR

Univariate OLS with intercept; HAC/Newey–West with 3 lags; minimum 18 observations. Currency-panel regressions cluster by month, require 80 observations, and add currency fixed effects plus EM and carry-rank interactions.

OUTCOME DEFINITIONS

$r_{t+1} = \sum_i w_{i,t} R_{i,t+1}$; long-leg and short-position PnL are the same sum restricted to positive and negative weights. Magnitude outcomes are $|r_{t+1}|$ and r_{t+1}^2 ; tail outcomes are $\mathbf{1}\{r_{t+1} < 0\}$ and $\mathbf{1}\{r_{t+1} \leq q_t^{5\%}\}$, where $q_t^{5\%}$ is the trailing 60-month, one-month-lagged 5th percentile of the same series.

UNITS

Return coefficients in basis points, probability coefficients in percentage points. The standardised economic coefficient multiplies the raw slope by the regression-sample standard deviation of the predictor; event-indicator regressions are not standardised and are reported as active-minus-inactive months.

EVIDENCE LABELS

Supported: expected sign in discovery and validation, valid inference in both, discovery $p < 0.10$ or BH $q < 0.20$. *Directionally suggestive*: expected sign without the significance condition. *Invalid inference* and *insufficient observations* are recorded separately, never silently merged into “no evidence”.

EXPECTED SIGNS

With risk sign ρ_j , the expected continuous coefficient is $-\rho_j$ for the mean outcome and $+\rho_j$ for magnitude and tail: more risk should mean lower expected return, larger moves and a higher severe-loss probability. Event regressions always expect $-1, +1, +1$.

DIAGNOSTICS CARRIED EXPLICITLY

36 regressions skipped for insufficient variation; 2 produced an invalid covariance matrix and were flagged rather than reported; the standardisation invariance audit passes on all 1,880 continuous regressions; 25 event regressions are excluded from standardisation by construction.

A3 — Frozen-evaluation results for the strategies discussed

Strategy	Ret.	Vol.	Sharpe	Δ SR vs base	Max DD	Gross	Tier
Baseline	+10.95%	6.16%	+1.78	+0.000	-6.61%	2.00	Reference / not ranked
A	+12.24%	6.25%	+1.96	+0.180	-6.85%	2.00	Tier 2 promising candidate
B	+10.59%	5.66%	+1.87	+0.090	-6.36%	2.00	Tier 2 promising candidate
C	+5.69%	3.47%	+1.64	-0.140	-3.65%	1.15	Tier 3 exploratory watchlist
A+B	+9.85%	5.81%	+1.69	-0.085	-6.08%	2.00	Tier 3 exploratory watchlist
A+C	+6.56%	3.38%	+1.94	+0.161	-3.85%	1.13	Tier 2 promising candidate
OIL 50\% overlay	+8.93%	5.62%	+1.59	-0.190	-6.61%	1.79	Tier 3 exploratory watchlist

READING THIS TABLE

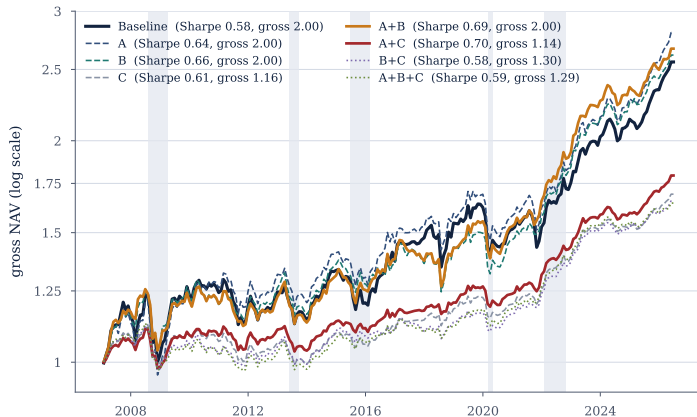
47 months, August 2022 to June 2026. The Δ column is against the *baseline over the same months*, the only comparison that removes the period effect. A 47-month Sharpe has a standard error near $\sqrt{12/47} \approx 0.5$, so none of these differences is distinguishable from zero.

WHY THE PERIOD WAS KIND TO CARRY

The window opens after the 2022 dollar and rates shock and contains no comparable episode. A book long EM against low-yield funders does well in exactly that environment, which is why baseline posts 1.78 against its long-run 0.58.

A4 — Core architecture NAV

Gross NAV before costs; shaded bands are the five fixed crisis windows



WHAT THE PATHS SHOW

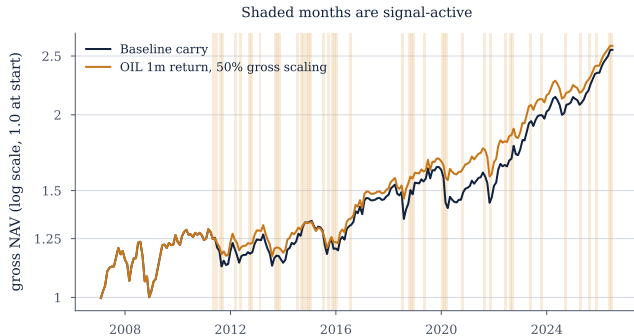
The four fixed-gross books share a shape because they hold overlapping baskets at the same size, separating gradually through selection. The four containing C run visibly flatter with shallower draw-downs: not a different strategy so much as a smaller one.

CRISIS BANDS

Shaded regions are the five fixed crisis windows. The 2008–09 and 2020 bands are where the gross-controlled books earn their keep.

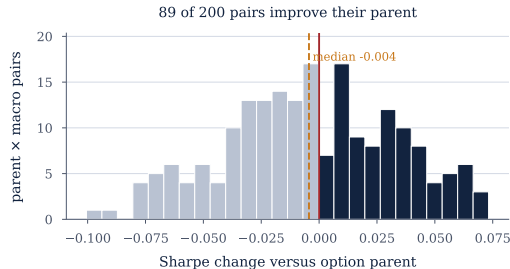
Gross of costs. Streams rebuilt from the Notebook 07 and 08 weight files and verified to reproduce 08v6_core_option_comparison.parquet exactly.

A5 — Oil overlay NAV, and the distribution of incremental effects



READING THE PATHS

The two are indistinguishable for long stretches and separate inside the shaded clusters. Note 2008–09: the oil signal was mostly *not* active through the GFC, which is why the overlay's maximum drawdown equals the baseline's.



READING THE HISTOGRAM

The 200 parent-feature Sharpe changes are centred slightly below zero; dark bars are the 89 improvements. This is what a genuine but weak, feature-specific effect looks like — a systematic effect would sit clearly to the right of zero.

A6 — Transaction-cost sensitivity

Net gross-Sharpe under a linear roll-plus-rebalancing cost

Strategy	Gross	1 bp	2 bp	5 bp	10 bp	25 bp	50 bp
Baseline	+0.58	+0.55	+0.52	+0.43	+0.29	-0.14	-0.86
A	+0.64	+0.61	+0.58	+0.49	+0.34	-0.11	-0.86
B	+0.66	+0.62	+0.59	+0.49	+0.31	-0.20	-1.06
A+B	+0.69	+0.65	+0.62	+0.51	+0.33	-0.21	-1.11
A+C	+0.70	+0.66	+0.63	+0.52	+0.33	-0.21	-1.13
OIL 50\% overlay	+0.65	+0.62	+0.59	+0.50	+0.34	-0.13	-0.90

COST MODEL

Net return subtracts a **roll cost** on gross notional $\sum_i |w_{i,t}|$ and a **rebalancing cost** on one-way turnover $\frac{1}{2} \sum_i |w_{i,t} - w_{i,t-1}|$, both at the same per-unit rate. A variable-gross book therefore pays less roll cost automatically.

WHAT IT IMPLIES

Ordering is stable at realistic 1–5 bp levels. At 25–50 bp the higher-turnover selection books lose their advantage first, while gross-controlled books degrade more slowly because both cost terms are smaller. Cost is part of the mechanism story, not a separate footnote.

A7 — Strategy universe and research-confidence taxonomy

Strategy family	Canonical strategies
macro extension	161
standalone macro	46
ultimate exploratory matrix	35
parent-relative tilt	16
core option	8
composite macro	5
Total	271

DE-DUPLICATION

323 named strategies collapse to **271** canonical return streams after removing 52 aliases with numerically identical weights. Counting aliases would inflate every breadth statistic in the study.

Research-confidence tier	Count	Reading
Tier 1 robust shortlist	0	passes every screen simultaneously
Tier 2 promising candidate	98	survives several screens; not independently validated
Tier 3 exploratory watchlist	172	retained for the record only
Reference / not ranked	1	Baseline carry, the comparison benchmark

WHAT THE TIERS ARE NOT

A **research-confidence taxonomy**, not a count of validated alphas. Tier 2 does not mean 98 profitable strategies; it means 98 streams cleared several screens and none cleared all of them. The empty Tier 1 is the intended output of a strict screen.

WHY 271 IS NOT 271 HYPOTHESES

Strategies share parents, share signals, and sometimes differ only in a scaling constant. The effective number of independent tests is far below 270.

A7b — Point-in-time verification status

All 271 canonical strategies carry a complete lineage row; the status records how far verification could go

Point-in-time status	Strat.	What it means
calendar date verified, intraday unverified	120	signal dates precede the return window; intraday timestamps unavailable
source logic supported, not independently rebuilt	96	construction logic documented and inherited, selection not re-derived here
insufficient source information	50	source metadata too thin to verify beyond lineage
return timing verified, no conditioning signal	5	the strategy has no conditioning signal to audit
Total	271	complete lineage for all; verification evidence complete for 221, incomplete for 50

THE FINDING

Zero future-horizon violations in the macro, option and strategy-level audits: no signal used to form a position at month t draws on information dated after t .

THE LIMITATION

Absence of detected leakage is not proof of correctness. 120 strategies are verified at calendar-date resolution only, and 146 could not be independently re-derived from the stored metadata.

A7c — Fixed crisis windows

Total return over each window, and average gross exposure across all five

Strategy	GFC 2008-09	Taper 2013	EM/China 2015-16	Covid 2020	USD/Rates 2022	Avg. gross
Baseline	-9.18%	-4.85%	-4.64%	-7.18%	+12.73%	2.00
A	-10.29%	-4.85%	-3.74%	-6.79%	+12.32%	2.00
B	-9.37%	-5.88%	-1.76%	-6.29%	+13.77%	2.00
A+B	-11.38%	-5.88%	-0.45%	-5.37%	+14.09%	2.00
A+C	-6.80%	-3.84%	-1.23%	-3.10%	+8.08%	1.11
OIL 50\% overlay	-9.18%	-5.26%	-1.51%	-1.45%	+9.98%	1.66

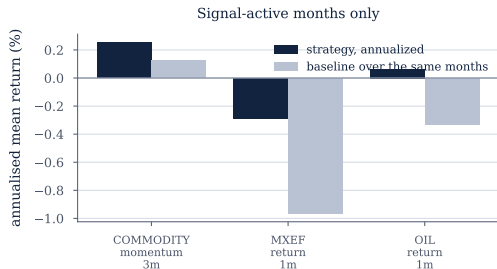
WINDOW DEFINITIONS

GFC 2007-07 to 2009-03; taper tantrum 2013-05 to 2013-09; EM and China stress 2015-06 to 2016-02; Covid 2020-02 to 2020-04; dollar and rates shock 2022-01 to 2022-10. Chosen ex post from market history, so these are a stress description, not a test.

WHAT TO LOOK FOR

Compare each row against baseline in the same column. Fixed-gross selection books track baseline closely; the gross-controlled ones diverge, and they diverge by different amounts in different episodes. Average gross is the column that explains most of the difference.

A8 — Signal-active months



RULE-DEFINED PERIODS

Each bar covers only the months in which that strategy's own signal was active, alongside baseline over the identical months. This tests the rule rather than the calendar.

WHAT IT CONFIRMS

Active months are, on average, *not* losing months for baseline carry. The overlays run half-sized through a mix of good and bad months — exactly the picture the 41.3% precision statistic describes for oil. Any protective value must therefore be judged conditionally, which is why slide 11 splits active months into baseline-loss and baseline-gain sub-samples rather than averaging over them.

08v6_risk_period_results.parquet, period type "rule-defined active months"; 17,615 rows across six strategy families.

A9 — Presentation fact-check register (1 of 3)

Every main-deck number, its source object, and whether the build could verify it

Claim	Value used	Notebook source object	Source row	Ver.
Full sample 2007-01-31 to 2026-06-30, 234 months	2007-01-31 to 2026-06-30, 234 months	08v6_summary.json	full_sample_dates / full_sample_months	yes
Discovery 2007-01-31 to 2018-08-31, 140 months	['2007-01-31', '2018-08-31'], 140	08v6_summary.json	discovery_dates / discovery_months	yes
Validation 2018-09-30 to 2022-07-31, 47 months	['2018-09-30', '2022-07-31'], 47	08v6_summary.json	validation_dates / validation_months	yes
Evaluation 2022-08-31 to 2026-06-30, 47 months	['2022-08-31', '2026-06-30'], 47	08v6_summary.json	evaluation_dates / evaluation_months	yes
25 point-in-time macro features	25	08v6_macro_feature_audit.parquet	row count	yes
3 supported mean-return features	MXEF_return_1m, MXEF_tail_return_1m, COMMODITY_momentum_3m	08v6_summary.json	supported_mean_features	yes
4 supported magnitude features	VIX_level, JPMVXYEM_change_1m, MXEF_momentum_3m, MXEF_drawdown_6m	08v6_summary.json	supported_magnitude_features	yes
8 supported tail-risk features	JPMVXYEM_change_1m, MXEF_return_1m, MXEF_momentum_3m, MXEF_drawdown_6m, MXEF_tail_return_1m, COMMODITY_return_1m, COMMODITY_momentum_3m, OIL_return_1m	08v6_summary.json	supported_tail_features	yes
14 explicit predictive/contemporaneous matched pairs	14	08v6_summary.json	lead_concurrent_explicit_matched_pair_count	yes
Baseline full-sample Sharpe 0.58, max drawdown -19.0%	0.5779, -0.1900	08v6_core_option_comparison.parquet	Baseline carry	yes
A+B full-sample Sharpe 0.69, average gross 2.0	0.6880, 2.0000	08v6_core_option_comparison.parquet	Core A+B	yes
A+C full-sample Sharpe 0.70, average gross 1.14, MDD -11.6%	0.6989, 1.1351, -0.1157	08v6_core_option_comparison.parquet	Core A+C	yes

A9 — Presentation fact-check register (2 of 3)

Claim	Value used	Notebook source object	Source row	Ver.
Baseline frozen-evaluation Sharpe 1.78	1.7793	08v6_evaluation_leaderboard.parquet	Baseline carry	yes
A+B evaluation Sharpe 1.69, i.e. -0.09 versus evaluation baseline	1.6942, delta -0.0851	08v6_evaluation_leaderboard.parquet	Core A+B	yes
A+C evaluation Sharpe 1.94, i.e. +0.16 versus evaluation baseline	1.9399, delta +0.1606	08v6_evaluation_leaderboard.parquet	Core A+C	yes
OIL 50% overlay full-sample Sharpe 0.65	0.6511	08v6_summary.json	oil_50pct_overlay_summary.full_sample_sharpe	yes
OIL 50% overlay frozen-evaluation Sharpe 1.59	1.5896	08v6_summary.json	oil_50pct_overlay_summary.frozen_evaluation_sharpe	yes
OIL conditional protective benefit 16.3% annualized	16.30%	08v6_summary.json	oil_50pct_overlay_summary.protective_benefit	yes
OIL false-positive cost 12.03% annualized conditional	12.03%	08v6_summary.json	oil_50pct_overlay_summary.false_positive_cost	yes
OIL net active annualized delta -0.06%	-0.06%	08v6_summary.json	oil_50pct_overlay_summary.net_active_annualized_delta	yes
OIL event precision 41.3%	41.30%	08v6_summary.json	oil_50pct_overlay_summary.event_precision	yes
271 canonical strategies	271	08v6_summary.json	canonical_strategy_count	yes

A9 — Presentation fact-check register (3 of 3)

Claim	Value used	Notebook source object	Source row	Ver.
270 active non-reference multiple-testing hypotheses	270	08v6_summary.json	active_non_reference_multiple_testing_count	yes
Tier 1 = 0, Tier 2 = 98, Tier 3 = 172, Reference = 1	0, 98, 172, 1	08v6_summary.json + 08v6_full_strategy_classification_registry.parquet	robust_shortlist_rows / promising_candidate_list_rows / exploratory_watchlist_rows	yes
PIT status: 120 / 96 / 50 / 5	calendar_date_verified_intraday_unverified=120, source_logic_supported_but_not_independently_reconstructed=96, insufficient_source_information=50, return_timing_verified_no_conditioning_signal=5	08v6_summary.json	strategy_point_in_time_status_counts	yes
All 271 strategies have complete PIT lineage; evidence complete for 221, incomplete for 50	lineage true=271, evidence true=221, false=50	08v6_summary.json	strategy_pit_lineage_rows_complete_counts / ..._evidence_complete_counts	yes
Zero future-horizon leakage detected	0	08v6_summary.json	lookahead_violation_count / macro_pit_reconciliation_future_horizon_violations	yes
200 parent x macro combinations: 89 raise and 111 lower parent Sharpe	200 rows, 89 positive, 111 negative	08v6_macro_extension_incremental_results.parquet	Delta Sharpe versus parent	yes
OIL overlay on baseline adds +0.073 Sharpe; on A+C it adds +0.028	baseline +0.073, A+C +0.028	08v6_macro_extension_incremental_results.parquet	Baseline carry / Core A+C x OIL_return_1m	yes
A+C + OIL cuts average gross from 1.14 to 1.02	1.1351 to 1.0224	08v6_macro_extension_incremental_results.parquet	Core A+C x OIL_return_1m average gross	yes
No strategy's active return survives BH correction (0 of 270 with BH q<0.10)	BH<0.10: 0; raw HAC p<0.05: 36	08v6_gross_statistical_tests.parquet	BH_adjusted_p_value	yes

Generated by `scripts/build_assets.py`, which fails the build if any claim cannot be verified against the notebook's saved outputs. Machine-readable version:

`build/presentation_fact_check_table.csv`; full extracted fact set: `build/facts.json`.

A10 — Limitations

STATISTICAL

No strategy survives BH correction. The evaluation window is 47 months. Univariate regressions identify association, not causation, and the 25 features are strongly cross-correlated, so the count of supported features overstates the number of independent findings.

CONSTRUCTION

Two features are numerically duplicated under different labels. The 50% multiplier and the 80th/20th threshold are fixed by judgement rather than estimated — this avoids in-sample fitting but leaves the rule un-optimised. Composite breadth is feature-level, not asset-class-level.

VERIFICATION

50 strategies have insufficient source information for verification beyond lineage, and the four combined option architectures inherit selection logic that is documented but not independently re-derived here.

ECONOMIC

Sharpe ratios are gross of the risk-free rate: annualised arithmetic return over annualised volatility, with no financing subtracted. Costs are a linear per-unit charge on roll notional and turnover, with no market impact or capacity constraint. Variable-gross results are not leverage-matched to baseline.